Kevin Shen

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Education

University of Waterloo Sept 2025 – Expected Jun 2030

Bachelor of Applied Science in Mechatronics Engineering

- **Class Representative** of **117** Students, Mechatronics Class of 2030
- **Student Life Director**, Engineering Society

Experience

Propulsion & Trailer Ground Equipment Member — *Waterloo Rocketry* Sept 2025 – Present

- Designed and machined **custom stainless-steel plumbing adapters** (BSPP–NPT) in **SolidWorks** using industry standard machining tools to correct tolerances
- Executed **2 cold-flow tests** involving integrating plumbing, instrumentation setup, and data logging to validate flow feed and valve behaviour
- **Co-developed preliminary design for adjustable oxidizer tank tower**, targeting safety and precision of an 80 kg propellant tank

Founder & Detailing Technician — *KevCleans (Mobile Auto Detailing)* May 2025 – Sept 2025

- Founded and scaled a mobile service to **$2,000+ revenue** using standardized workflows and pricing
- Trained and supervised a detailing technician on equipment use, chemical safety, and service procedures, maintaining a **90%+** repeat and referral rate
- Performed **30+** interior and exterior detailing using professional tools and chemicals, including pressure washers, foam cannons, extractors, and polishers

Land Surveyor Assistant — *Delph & Jenkins Land Surveyors* Feb 2024 – Jun 2025

- Processed and archived **100+** legal survey documents and reference plans, verifying dimensions and boundaries
- Produced **10+ AutoCAD drawings** and grading plans supporting subdivision and site submissions
- Reduced revision cycles by validating boundaries against legal descriptions prior to final issue

Math Instructor — *Mathnasium Learning Centre* Aug 2024 – Jun 2025

- Instructed and coached **50+ students per week** on elementary to high-school math curriculum, adapting learning strategies to each student
- Contributed to an average **∼20% improvement in test and assessment scores over a 4-month period** by delivering targeted instruction and structured problem-solving strategies

Projects

Intelligent Fall Detection System — *Wearable Hardware + iOS Mobile App*

- Built an **ESP32-based wearable** using **Arduino** accelerometer to detect falls from seniors, prototyping across **50+** simulated fall and non-fall scenarios
- Developed a **Flutter iOS application** in **Xcode**, for accessible usage and connection to other devices
- Implemented low-latency **Bluetooth Low Energy** communication from belt to app, enabling instant detection and assistance

Rover Prototype — *Waterloo Engineering Competition*

- Designed an auto-balancing rover using **Arduino UNO**, DC motors, and custom traction tires
- Iterated **three** prototypes in 7 hours to achieve autonomous balance on a pivoting beam with 3 team members

Self-Made Engineering Ring — *Manual Machining Project*

- Measured finger dimensions and machined a **hollow metal ring** on a **manual metal lathe** to precise fit with 0.5 mm tolerance
- Cut and shaped surface notches using **vice grips and hand filing**, maintaining symmetry and consistent spacing
- Finished the ring through **polishing** to achieve a smooth, wearable surface

Skills

CAD & Mechanical: SolidWorks, AutoCAD, Fusion 360, 3D Printing, Manual Machines
Embedded & Software: Arduino / ESP32, Python, C++, Java, Git/Github, Version Control